

AST1410 (Winter 2026) MESA Practical 3

Due 11:59pm 8 Mar 2026

Practicals must be submitted to essick@cita.utoronto.ca before the deadline.

Practicals should be submitted as a PDF describing your results.

Your PDF must be accompanied by a plain ASCII file with all scripts, etc that you developed or a link to a public github repository. If you choose to submit as a github repository, you must specify which commit/hash and branch you are submitting.

Part 1 (10 points)

Compute and plot the isochrones in an HR diagram for stars with solar metallicity. Identify the turn-off mass and plot it as a function of age.

Part 2 (5 points)

Using your results, estimate the age of the globular cluster shown in the right panel of Fig. 1.2 in the Pols lecture notes. Be sure to include an estimate of your uncertainty along with a point estimate of the age.